

Frédéric Andrian

Software Engineering · Backend, Full-Stack and AI

Bordeaux, France

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Professional Summary

Soon-to-be 2026 engineering graduate completing an MSc-equivalent degree in Artificial Intelligence and Cognitive Science, with hands-on experience across backend, full-stack development, AI, data, and deployment. At INRIA (French National Institute for Research in Digital Science and Technology), I independently delivered an application from a user need to a functional solution using Python, FastAPI, REST APIs, MongoDB, React, Next.js, Docker, and GitHub CI/CD. I have also developed academic projects with C#, .NET, Entity Framework Core, and Java. My research internships in France and India strengthened my autonomy, adaptability, and analytical approach. I want to contribute to varied software projects, continue learning across technologies, and grow within Netlight's international consulting environment. Available from November 2026 and willing to relocate to Amsterdam.

Experience

INRIA (French National Institute for Research in Digital Science and Technology), Mnemosyne Team

Bordeaux, France

AI Research Engineer Intern

Mar. 2026 – Sep. 2026

- Independently designed and developed, with guidance from my supervisors, an end-to-end application that helps researchers discover scientific papers.
- Translated a user need defined with my supervisor into a functional solution covering backend services, data persistence, AI components, and a web interface.
- Built Python and FastAPI backend services, REST APIs documented with OpenAPI, and MongoDB persistence.
- Developed a React and Next.js interface that enabled researchers to interact with the application and its tools.
- Integrated the OpenAI API and designed the tools used by a multi-step agent to retrieve information and complete successive tasks.
- Implemented a RAG pipeline using document chunking, embeddings, semantic search, and ChromaDB.
- Dockerized selected components and implemented a GitHub CI/CD pipeline to automate integration.
- Managed a Dockerized web application for LLMs installed locally on a Mac Studio, including model installation and configuration.
- Developed a Python tool to automatically collect scientific paper metadata from multiple public sources.
- Implemented webhooks to relay the results of scheduled tasks.
- Evaluated OpenClaw in an isolated Docker environment with attention to the reliability and security of autonomous tool and file access.
- Benchmarked Qwen, Gemma, and GLM and evaluated several quantization configurations on an HPC cluster with Slurm.
- Defined an experimental protocol, analyzed the results, and contributed to a scientific manuscript documenting the model evaluation.

Karunya University

Coimbatore, India

Reinforcement Learning Research Intern

Jun. 2025 – Sep. 2025

- Designed and compared deterministic path planning with a reinforcement learning approach for autonomous navigation of a hospital robot.
- Deployed and optimized the software and reinforcement learning model on an embedded Raspberry Pi.

Academic Project

University Management REST API

ENSC, Bordeaux INP

C#, ASP.NET Core, Entity Framework Core, SQLite

2024

- Developed a REST API in C# and ASP.NET Core to manage university student and course data.
- Implemented CRUD endpoints with controllers and DTOs, using Entity Framework Core and SQLite for persistence and migrations.
- Added Swagger/OpenAPI documentation to explore and interact with the API.

Technical Skills

Programming

Python, C#, .NET, ASP.NET Core, Entity Framework Core, Java, Node.js

Backend and APIs

FastAPI, REST APIs, Swagger/OpenAPI, MongoDB, SQLite, SQL, webhooks

Frontend

React, Next.js

Software Engineering

Linux, Docker, Git/GitHub, CI/CD, API design, full-stack development

AI Systems

LLMs, OpenAI API, multi-step agents, tool design, RAG, PyTorch, Deep Learning

Model Evaluation

Qwen, Gemma, GLM, quantization, experimental protocols, latency, TTFT, response quality

Languages

French (native), English (professional, IELTS 7.5), Spanish (basic)

Education

ENSC, Bordeaux INP

Bordeaux, France

Engineering Degree, MSc-equivalent, in Artificial Intelligence and Cognitive Science

2023 – 2026